Should a detergent line invest in material-loss controls?

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Recommendation: trial first, defer equipment commitment

The model supports further investigation, not capital approval. Validate the loss reduction on matched production runs and obtain delivered equipment quotations. At the current budget, process loss must fall from 3% to approximately 2.245% of input or lower to produce a non-negative five-year net present value (NPV).

Build the cost from the physical process

The hypothetical UK line sells 10 million one-litre bottles per year at a liquid density of 1.03 kg/litre. At 3% process loss, 10.619 million kg must enter the process to deliver 10.300 million kg of acceptable liquid. Reducing loss to 2% avoids approximately 108,353 kg of input annually at unchanged saleable output. Ingredient quantities are multiplied by assumed purchase rates; packaging, paid crew hours, electricity, disposal and freight are costed separately.

Investment measure	Modelled result
Production cost per bottle, including new upkeep and depreciation	GBP0.907 to GBP0.904
Annual gross avoided cost / net recurring cash saving	GBP57,205 / GBP49,205
Capital spending / total initial economic outflow	GBP92,000 / GBP117,767
Five-year NPV at a 10% discount rate	GBP46,392
Simple payback after start / including preparation	28.7 / 34.7 months
First-year operating profit effect	Reduction of GBP10,364

What could change the decision?

If loss falls only to 2.5%, five-year NPV becomes **negative GBP48,543**. The base-case equipment budget is GBP60,000; its modelled NPV-zero ceiling is approximately GBP100,341, with other assumptions unchanged. These thresholds are more useful for approval than a headline saving alone.

Costs, timing and evidence needed

Capital includes equipment, installation, validation and 15% contingency. Initial economic outflow also includes GBP5,000 pilot expense, GBP5,000 training and approximately GBP15,767 lost contribution from installation downtime. Savings begin in July; GBP8,000 annual maintenance is deducted. Existing payroll remains fixed. Project depreciation affects profit but is excluded from cash flow.

Before capital review: obtain calibrated batch input/output records, reject and rework data, quality results, current ingredient prices, equipment scope and quotations, metered energy impact, and an installation plan showing which lost output can be recovered. No factory measurements or commercial quotations have been supplied.

Source and limits

Unilever reports 1-2% savings in premium ingredients from dosing controls at its Cu Chi Home Care site. This supports investigating the mechanism; it does not validate the assumed 3%-to-2% total process-loss change. [1] All engineering quantities and purchase prices remain illustrative. This is a separate GBP line study, not a forecast of Unilever or a continuation of the earlier EUR100m teaching model.

[1] Unilever: digital twins across the manufacturing network (2026). Five-year pre-tax appraisal, no working-capital change or financing. Initial costs are conservatively charged at time zero; operating savings are discounted at year end.